

Integration by parts

Integration by parts (分部積分法) is a common technique for integration. It applies the product rule to integrals, transforming difficult-to-solve integral forms (especially those involving the multiplication of functions) into easier-to-solve integral forms.

First, the product rule for differentiation gives:

$$\frac{d}{dx}[f(x)\cdot g(x)] = g(x)\cdot\frac{d}{dx}f(x) + f(x)\cdot\frac{d}{dx}g(x)$$

Then integrate both sides of the equation. (integrate both L.H.S. and R.H.S.):

$$f(x)\cdot g(x) = \int\left[g(x)\cdot\frac{d}{dx}f(x) + f(x)\cdot\frac{d}{dx}g(x)\right]dx$$

$$f(x)\cdot g(x) = \int g(x)\cdot\frac{d\,f(x)}{dx}\,dx + \int f(x)\cdot\frac{d\,g(x)}{dx}\,dx$$

$$f(x)\cdot g(x) = \int g(x)\,d\,f(x) + \int f(x)\,d\,g(x)$$

And finally the formula for the method of integration by parts will be obtained:

$$\int f(x)\,d\,g(x) = f(x)\cdot g(x) - \int g(x)\,d\,f(x)$$

where C is integration constant

When encountering functions that cannot be integrated directly, by substitution, or using trigonometric identities (especially those involving function multiplication), students may attempt integration by parts.

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Integration by parts

Try it: What method should be used to integrate the following function? Indicate your choice with "✓".

Function	Direct Integration	Integration by substitution	Integration by Trigo. identities	Integration by parts
$\int x^{100}(2x+1)\,dx$				
$\int x(2x+1)^{100}\,dx$				
$\int \frac{x}{x^2+1}\,dx$				
$\int \frac{1}{x^2+1}\,dx$				
$\int \sin 2x\,dx$				
$\int \frac{1}{\sin^2 x}\,dx$				
$\int \sin^2 x\,dx$				
$\int \sin^3 x \cos^4 x\,dx$				
$\int xe^{2x}\,dx$				
$\int x \ln x\,dx$				
$\int x^2 \sin x\,dx$				
$\int e^x \sin x\,dx$				

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Integration by parts

e.g.1 Find $\int xe^{2x}\,dx$.

By consider $\int f(x)\,d\,g(x) = f(x)\cdot g(x) - \int g(x)\,d\,f(x)$

Let $f(x)=x$,

and $d\,g(x)=e^{2x}dx$

then $\frac{d}{dx}g(x)=e^{2x}$

$g(x)=\frac{1}{2}e^{2x}$

Therefore

$\int xe^{2x}\,dx$

$=\int x\,d(\frac{1}{2}e^{2x})$

$=x\cdot(\frac{1}{2}e^{2x})-\int(\frac{1}{2}e^{2x})\,dx$

$=\frac{1}{2}xe^{2x}-\frac{1}{4}e^{2x}+C$

Let $f(x)=e^{2x}$,

and $d\,g(x)=x\,dx$

then $\frac{d}{dx}g(x)=x$

$g(x)=\frac{x^2}{2}$

Therefore

$\int xe^{2x}\,dx$

$=\int e^{2x}\,d(\frac{x^2}{2})$

$=e^{2x}\cdot(\frac{x^2}{2})-\int(\frac{x^2}{2})\,de^{2x}$

$=$ .....

FAIL!

Let's TRY IT!

[M2 2014 Q.6(a)]

Find $\int xe^{-x}dx$. ***(3 marks)***

Integration by parts

e.g.2 Find $\int x \ln x \, dx$.

By consider $\int f(x)\, d\, g(x) = f(x)\cdot g(x) - \int g(x)\, d\, f(x)$

Let $f(x) = x$,
and $d\, g(x) = \ln x \, dx$
then $\frac{d}{dx} g(x) = \ln x$

$g(x) = ?$



FAIL!

Let $f(x) = \ln x$,
and $d\, g(x) = x \, dx$
then $\frac{d}{dx} g(x) = x$

$g(x) = \frac{x^2}{2}$

$\int x \ln x \, dx$

$= \int \ln x \, d(\frac{x^2}{2})$

$= \ln x \cdot (\frac{x^2}{2}) - \int (\frac{x^2}{2})\, d(\ln x)$

$= \frac{1}{2}x^2 \ln x - \int (\frac{x^2}{2}) \frac{1}{x} dx$

$= \frac{1}{2}x^2 \ln x - \int \frac{x}{2} dx$

$= \frac{1}{2}x^2 \ln x - \frac{x^2}{4} + C$

$\frac{d}{dx} \ln x = \frac{1}{x}$
$\therefore d \ln x = \frac{1}{x} dx$

[M2 2015 Q.4(a)]

Using integration by parts, find , find $\int x^2 \ln x \, dx$. ***(3 marks)***

Integration by parts

e.g.3 Find $\int x\sin x\,dx$.

By consider $\int f(x)\,d\,g(x) = f(x)\cdot g(x) - \int g(x)\,d\,f(x)$

Let $f(x) = x$,
and $d\,g(x) = \sin x\,dx$
then $\frac{d}{dx}g(x) = \sin x$

$g(x) = -\cos x$

Therefore

$\int x\sin x\,dx$
$= \int x\,d(-\cos x)$
$= x\cdot(-\cos x) - \int(-\cos x)\,dx$
$= -x\cos x + \sin x + C$

✓

Let $f(x) = \sin x$,
and $d\,g(x) = x\,dx$
then $\frac{d}{dx}g(x) = x$

$g(x) = \frac{x^2}{2}$

Therefore

$\int x\sin x\,dx$
$= \int \sin x\,d(\frac{x^2}{2})$
$= \sin x\cdot(\frac{x^2}{2}) - \int(\frac{x^2}{2})\,d(\sin x)$
$= \sin x\cdot(\frac{x^2}{2}) - \int(\frac{x^2}{2})\cos x\,dx$
$= .....$

FAIL!

$\frac{d}{dx}\sin x = \cos x$
$\therefore d\sin x = \cos x\,dx$

Let's TRY IT!

[M2 2012 Q.9(a)]

Using integration by parts, find $\int x\sin x\,dx$. ***(2 marks)***

Integration by parts (Double Strike)

Some more complex functions may require applying integration by parts twice consecutively to complete the integration:

e.g.4 Find $\int x^2 \cos x \, dx$.

Let $f(x) = x^2$,

and $d\,g(x) = \cos x \, dx$

then $\dfrac{d}{dx} g(x) = \cos x$

$g(x) = \sin x$

Therefore

$$\int x^2 \cos x \, dx$$
$$= \int x^2 \, d(\sin x)$$
$$= x^2 \cdot \sin x - \int (\sin x) \, d(x^2)$$
$$= x^2 \sin x - \int (\sin x) \, 2x \, dx$$
$$= x^2 \sin x - 2\int x \sin x \, dx$$

第一次分部積分開始！

完成後答案有 $\int x \sin x \, dx$，

那就要做第二次分部積分~

Consider $\int x \sin x \, dx$,

$$\int x \sin x \, dx$$
$$= \int x \, d(-\cos x)$$
$$= x \cdot (-\cos x) - \int (-\cos x) \, dx$$
$$= -x \cos x + \sin x + C$$

第二次分部積分開始 !!

完成後答案已是最簡，

不需再作更多積分了~

$$\int x^2 \cos x \, dx$$
$$= x^2 \sin x - 2\int x \sin x \, dx$$
$$= x^2 \sin x - 2(-x \cos x + \sin x + C)$$
$$= x^2 \sin x + 2x \cos x - 2 \sin x + C_1$$

將第一次及第二次分部積分的結果合併，大功告成

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Integration by parts (Double Strike)

[M2 2017 Q.4(a)]

Using integration by parts, find $\int x^2 e^{-x}\, dx$. ***(3 marks)***

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Integration by parts (Recursive)



Some more complex functions yield their own integral when integrated by parts (haunting you like a ghost)!



e.g.5 Find $\int e^x \cos x \, dx$.

Let $f(x) = e^x$,

and $d\,g(x) = \cos x \, dx$

then $\dfrac{d}{dx} g(x) = \cos x \quad \therefore \; g(x) = \sin x$

Therefore

$$\int e^x \cos x \, dx$$

$$= \int e^x \, d(\sin x)$$

$$= e^x \cdot \sin x - \int (\sin x) \, d(e^x)$$

$$= e^x \sin x - \int (\sin x) \, e^x dx$$

$$= e^x \sin x - \int e^x \sin x \, dx$$



第一次分部積分開始！

完成後答案有 $\int e^x \sin x \, dx$，

那就要做第二次分部積分~

Consider $\int e^x \sin x \, dx$,



$$\int e^x \sin x \, dx$$

$$= \int e^x \, d(-\cos x)$$

$$= e^x \cdot (-\cos x) - \int (-\cos x) \, d(e^x)$$

$$= -e^x \cos x + \int (\cos x) \, e^x dx$$

$$= -e^x \cos x + \int e^x \cos x \, dx$$

第二次分部積分開始 !!

完成後答案有 $\int e^x \cos x \, dx$，

返回原本要積分的算式…

$$\therefore \int e^x \cos x \, dx = e^x \sin x - \int e^x \sin x \, dx$$

$$= e^x \sin x - (-e^x \cos x + \int e^x \cos x \, dx)$$

$$= e^x \sin x + e^x \cos x - \int e^x \cos x \, dx$$

$$2\int e^x \cos x \, dx = e^x \sin x + e^x \cos x + C$$

$$\int e^x \cos x \, dx = \frac{1}{2} e^x (\sin x + \cos x) + C_1$$

將第一次及第二次分部積分的結果合併，大功告成

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Integration by parts (Recursive)



[P.Math 2007 Paper II Q.4(a)]

Find $\int e^x \sin x \, dx$. ***(4 marks)***

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D.I. Method

The DI method, also known as the tabular method, is a shortcut for integration by parts that organizes **repeated Differentiation and Integration** into a table.

It is especially useful for integrating a product of two functions where **one part can be differentiated until it becomes zero (like a polynomial)**.

e.g. Find $\int x^2 \cos x \, dx$.

By using DI Method:

Sign	D	I	Result
+	x^2	$\cos x$	
–	$2x$	$\sin x$	$+x^2 \sin x$
+	2	$-\cos x$	$-2x(-\cos x)$
–	0	$-\sin x$	$+2(-\sin x)$

$\therefore \int x^2 \cos x \, dx = x^2 \sin x + 2x \cos x - 2 \sin x + C$

e.g. Find $\int x^2 e^{-x} \, dx$.

By using DI Method:

Sign	D	I	Result
+	x^2	e^{-x}	
–	$2x$	$-e^{-x}$	$+x^2(-e^{-x})$
+	2	e^{-x}	$-2x\, e^{-x}$
–	0	$-e^{-x}$	$+2(-e^{-x})$

$\therefore \int x^2 e^{-x} \, dx = -x^2 e^{-x} - 2xe^{-x} - 2e^{-x} + C$

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D.I. Method

If both parts of the integral cannot be differentiated until it becomes zero (e.g. $\int e^x \cos x \, dx$), then the result may be recursive and it needs to pay more attention when using DI method.

e.g. Find $\int e^x \cos x \, dx$.

By using DI Method:

Sign	D	I	Result
+	e^x	$\cos x$	
−	e^x	$\sin x$	$+e^x \sin x$
+	e^x	$-\cos x$	$-e^x \ (-\cos x)$
$+ \int e^x(-\cos x)\, dx$			

or

Sign	D	I	Result
+	$\cos x$	e^x	
−	$-\sin x$	e^x	$+(\cos x)\ e^x$
+	$-\cos x$	e^x	$-(-\sin x)\ e^x$
$+ \int (-\cos x)\, e^x dx$			

Therefore,

$$\int e^x \cos x \, dx = e^x \sin x + e^x \cos x + \int e^x(-\cos x)\, dx$$

$$2\int e^x \cos x \, dx = e^x \sin x + e^x \cos x + C$$

$$\int e^x \cos x \, dx = \frac{e^x}{2}(\sin x + \cos x) + C_1$$

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D.I. Method

For integrals consisting of logarithm (e.g. $\int x^2 \ln x \, dx$), we must put the logarithm part into Differentiation (since logarithm cannot be integrated directly), and it needs to pay more attention when using DI method.

e.g. Find $\int x^2 \ln x \, dx$.

Note that sometimes DI Method may fail with inappropriate setup:

Sign	D	I	Result
+	x^2	$\ln x$	
−	$2x$	?	?

By using DI Method with appropriate setup:

Sign	D	I	Result
+	$\ln x$	x^2	
−	$\frac{1}{x}$	$\frac{x^3}{3}$	$+\frac{x^3}{3}\ln x$
$-\int \frac{1}{x}\cdot\frac{x^3}{3}\,dx$			

Therefore,

$$\begin{aligned}\int x^2 \ln x \, dx &= \frac{x^3}{3}\ln x - \frac{1}{3}\int x^2 \, dx \\ &= \frac{x^3}{3}\ln x - \frac{x^3}{9} + C\end{aligned}$$

Note:

DI Moethod is only suitable for dealing with certain types of integrals such as $\int x^2 \cos x \, dx$ and $\int x^3 e^x \, dx$, and you may risk losing M marks if you apply this method in the HKDSE.

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HKAL Questions (Level 1)

1. ***[P.Math 1992 Paper II Q.4 Modified]***

Find $\int xe^{x-1}dx$. ***(4 marks)***

2. ***[P.Math 2000 Paper II Q.1]***

Find $\int x\cos x\,dx$. ***(2 marks)***

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HKAL Questions (Level 1)

3. ***[P.Math 1998 Paper II Q.3]***

Find $\int \ln(1+x)\,dx$. ***(3 marks)***

4. ***[P.Math 2002 Paper II Q.9(a)]***

Find $\int e^{-x}\sin x\,dx$. ***(3 marks)***

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HKDSE Questions (level 1)

5. ***[M2 2021 Q.7(a)]***

Using integration by parts, find $\int (\ln x)^2 \, dx$. ***(3 marks)***

6. ***[M2 2018 Q.4(a)]***

Using integration by parts, find $\int u\,(5^u)\, du$. ***(3 marks)***

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HKAL Questions (Level 2)

7. ***[P.Math 1996 Paper II Q.5(a)]***

(a) Find $\int x \ln x \, dx$.

(b) Find $\int \frac{\ln x}{x} \, dx$. ***(6 marks)***

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HKAL Questions (level 2)

8. ***[P.Math 2008 Paper II Q.4(a)]***

(i) Find $\int \frac{1}{1+4x^2}\,dx$.

(ii) Hence find $\int \ln(1+4x^2)\,dx$. ***(4 marks)***

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HKAL Questions (level 2)

9. ***[P.Math 1993 Paper II Q.5]***

Find $\int e^{2x}(\sin x + \cos x)^2 \, dx$. ***(7 marks)***

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HKAL Questions (Level 3)

10. ***[P.Math 1990 Paper II Q.7(a)]***

Find $\int \ln(1+x^2)\,dx$. ***(4 marks)***

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HKDSE Questions (level 3)

11. ***[M2 2024 Q.6(a)]***

Let k be a constant. Using integration by parts, prove that

$$\int \cos(k\ln x)\,dx = \frac{x}{1+k^2}\left(\cos(k\ln x) + k\sin(k\ln x)\right) + \text{constant}\,. \qquad \textbf{(3 marks)}$$

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HKAL Questions (Level 3)

12. ***[P.Math 1997 Paper II Q.1]***

(a) Show that $\frac{d}{dx}\tan\frac{x}{2}=\frac{1}{1+\cos x}$.

(b) Using (a), or otherwise, find $\int\frac{x+\sin x}{1+\cos x}\,dx$. ***(5 marks)***

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HKAL Questions (Level 4)

13. ***[P.Math 2001 Paper II Q.2 Modified]***

(a) Find $\int \frac{1}{1+x^2}\,dx$.

(b) Hence find $\int x^2 \tan^{-1} x\,dx$. ***(6 marks)***

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HKAL Questions (Level 4)

14. ***[P.Math 2005 Paper II Q.4 Modified]***

(a) Find $\int \sqrt{1+x}\, dx$.

(b) Find $\int \frac{1}{\sqrt{1-x^2}}\, dx$.

(c) Using (a) and (b), find $\int \frac{\sin^{-1} x}{\sqrt{1-x}}\, dx$. ***(6 marks)***

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